Supplementary Material

Layer-wise Stage-aware Adapter Scaling for Multi-view Diffusion Texture Generation

1 S1. Frozen protocols and provenance

The main clean-v2 evaluation uses the controlled-retraining checkpoint with SHA-256:

`0618d6b284ab47aa16d7a89dc447f5ba`
`4455ff6c9d918e5e129478a99e0114c0`

The strict holdout contains 276 objects and has ordered-list SHA-256:

`a6aa8ab6e475763e1b5e67dc8c712ec8`
`f1940e3952d65897c820887554bbd044`

All image evaluations use six unique views $(0, 15, 12, 16, 13, 14)$, 50 denoising steps, seed 42, and object-level paired bootstrap with 10,000 resamples and seed 20260928. The main four-condition run and the stage-placement follow-up are separate protocol surfaces. Their repeated C3 outputs are not identical and are therefore not pooled.

The complete records/CSV/PNG/hash audit is in `coordination/E0A_EVIDENCE_FREEZE_20260929.json`. It verifies 276 record JSON files, 1,104 stage rows, 1,104 serialized Full-SSIM rows, and exact object–condition pairing. The follow-up uses early/middle/late step counts 17/16/17. Nominal scale summaries are:

Table 1: Nominal and recorded budget quantities for the stage-placement follow-up.

Schedule	Step mean	Scale sum	Squared-scale sum	Actual L2 mean
Fixed mean	1.6667	83.3333	138.8889	8936.92
HLL	1.6750	83.7500	157.8125	9239.76
LHL	1.6500	82.5000	153.1250	8799.86
LLH	1.6750	83.7500	157.8125	8777.53

The residual norms are descriptive measures of effective intervention intensity; they do not identify residual-direction alignment or establish a causal mechanism.

2 S2. Complete stage-placement metrics

The follow-up means are reproduced below. LPIPS is lower-is-better. The six non-SSIM metrics all favor LLH over LHL under their intended directions.

Schedule	Full-PSNR	Full-SSIM (pre-save)	Full-LPIPS	FG-PSNR	FG-SSIM	FG-LPIPS	Edge-SSIM
Fixed mean	14.311	0.8435	0.188	6.257	0.316	0.205	0.499
HLL	13.397	0.8252	0.202	5.583	0.259	0.212	0.464
LHL	14.868	0.8546	0.190	6.760	0.348	0.201	0.500
LLH	15.136	0.8518	0.181	6.937	0.357	0.200	0.536

The saved-artifact Full-SSIM means are fixed-mean 0.87395, HLL 0.85993, LHL 0.88144, and LLH 0.88313. LHL minus fixed-mean is +0.00749 ([0.00674,0.00826]), LHL minus HLL is +0.02151 ([0.01984,0.02319]), and LHL minus LLH is −0.00170 ([-0.00205,-0.00134]). These are paired object-level 95% bootstrap intervals, not equivalence tests.

3 S3. Fixed-GT Full-SSIM decomposition

The controlled 12-object trace saved A (pre-save prediction), B (PNG encoder input), C (PNG-reloaded prediction), the original float GT, and the PNG-reloaded GT. A and B are exactly equal. With the GT held fixed, the mean SSIM effects are:

This decomposition is diagnostic and does not recover historical 300-object pre-save tensors. The machine-readable rows are in `FIXED_GT_SSIM_DECOMPOSITION.csv/json` under the controlled trace directory.

Table 2: Mean Full-SSIM changes in the fixed-GT 12-object decomposition.

Comparison	Mean change
A fp16 → A fp32, original GT fixed	+0.03198
A fp32 → C fp32, original GT fixed	−0.00036
Original GT → PNG GT, A fp32 fixed	−0.00007
A/original GT → C/PNG GT	−0.00043

4 S4. Twelve-object Exact-GLB baking audit

The stratified case study contains 12 frozen Exact-GLB objects, four generated conditions, 48 exported textured GLBs, and 528 unseen-view rows (11 views per object–condition). Six source views were used for baking. The object-level means are:

Table 3: Unseen-view baking means for the selected case-study cohort.

Method	Masked PSNR	CIEDE2000	FG-LPIPS	Silhouette IoU
No adapter	11.824	18.200	0.156	0.99968
Fixed-low	6.001	48.623	0.199	0.99968
Fixed-high	3.891	60.420	0.216	0.99968
C3/LHL	5.638	51.183	0.201	0.99968

Coverage, post-inpainting coverage, seam definitions, and per-object rows are provided in the `cpu_bake_12` audit directory. DISTS was unavailable; the cohort has no generated 12-object GT bake. Blender reindexed vertices for `obj_0048`; this is not a byte-identity guarantee. Generated renders show source-fusion colour inconsistency, so this audit is implementation/case-study evidence, not a population-level TCAS advantage.

Table 4: Concrete object-level baking failure boundaries. Values are object means over 11 unseen views; paired entries are fixed-low/C3.

Object	Raw cov.	Inpaint frac.	Masked PSNR	CIEDE2000	FG-LPIPS	Boundary
`obj_0048`	0.1547	0.8441	5.090/4.927	53.328/54.339	0.388/0.375	Blender reindexing
`obj_0078`	0.0328	0.2802	4.278/3.879	58.822/61.431	0.066/0.068	very low raw coverage
`obj_0082`	0.0286	0.4951	4.144/3.823	55.558/57.364	0.081/0.082	very low raw coverage

These cases make the failure boundary concrete: the low-coverage objects remain limited even after propagation, while `obj_0048` obtains most of its final texture coverage from inpainting. The GT-to-GT sanity check was only a two-object smoke (`obj_0013` and `obj_0015`); it is not a 12-object GT bake. The bake handoff has blank canonical `source_asset_sha256` fields, and the saved seam scalar does not include a seam-pair count, so source identity and seam denominators remain audit limitations.

5 S4a. Complete-object and unseen-view index

Figure 1 provides the complete 12-object contact sheet used to select the qualitative panels in the main paper. Individual full-size PNG panels are retained under `main_adapter_clean_v2/comparisons/`, with one file per object and the columns GT, no adapter, fixed-low, fixed-high, and C3. The object IDs are the frozen stratified baking cohort and the same IDs are used by the object-level CSV rows in `cpu_bake_12`. The unseen-view directory contains 11 rendered views per object–condition; missing or low-coverage rows are retained as limited observations and are not converted to zero scores.

6 S4b. Layer-wise bake extension with same-draw global controls

The layer-wise bake uses the same 12 Exact-GLB objects, the same bake protocol (1024 texture resolution, 512 render resolution, 6 source views, 11 unseen views), and the same frozen handoff geometry/UV/cameras/GT as

S4. The layer-wise panels and fresh global fixed-low/C3 controls were generated by ONE seeded runner (per-object seeds fix the reference preprocessing draw, the initial latent, and the condition-VAE path), so the rows below are directly comparable; they are NOT comparable with the absolute values in S4, which come from an earlier runner with different reference draws.

Table 5: Same-draw unseen-view bake means over the 12-object cohort (layer-wise extension; 11 unseen views per object–condition).

Method	Masked PSNR↑	CIEDE2000↓	FG-LPIPS↓	Texture coverage
Global fixed-low (same-draw)	10.763	26.287	0.1705	0.1574
Global C3 (same-draw)	12.565	21.643	0.1562	0.1574
Layer-LHL	15.005	16.922	0.1420	0.1574
Layer-LLH	16.679	14.186	0.1215	0.1574

Layer-LLH beats layer-LHL on masked PSNR (12/12 objects), CIEDE2000 (12/12), and FG-LPIPS (12/12), and also beats the same-draw global C3 control (12/12 on all three); the global C3-vs-fixed-low ordering is preserved (12/12). The ordering holds on the failure-boundary objects `obj_0048` (14.97 vs 13.67 dB), `obj_0078` (12.02 vs 9.18 dB), and `obj_0082` (11.01 vs 8.63 dB). As in S4, this is a stratified case study: no generated GT bake, no DISTS, seam scalars without a seam-pair denominator, and the same inpainting rule for all conditions. The per-object and per-view rows are in `cpu_bake_12_layerwise/`.

7 S5. Generic schedule audit from the original protocol

The first-round request for generic schedules is answered with the retained strong-residual, per-layer-capped audit rather than by mixing it into the clean-v2 main table. It compares the frozen C3 schedule with a linear warm-up and a cosine bump on the original 276-object record. C3 minus each competitor is positive for PSNR in the paired bootstrap comparison, but the adapter instance, protocol, and metric path are distinct from the clean-v2 panel. These values support a bounded comparison of schedule families, not a new claim of cross-protocol transfer.

Table 6: Original-protocol generic schedule comparison.

Schedule	FG-SSIM	PSNR	ΔPSNR vs. C3	C3 wins
Linear warm-up	0.377	18.97	+0.323 [0.263,0.386]	214/276
Cosine bump	0.330	18.39	+0.902 [0.824,0.981]	253/276
C3 (TCAS)	0.376	19.29	–	–

The original record also contains trapezoid and Gaussian-peak controls; their complete rows remain in the internal evidence freeze. The linear warm-up is close in FG-SSIM but lower in PSNR, while the cosine bump is lower on both values. We retain this as evidence that TCAS is meaningfully comparable to common smooth schedules, without claiming optimality across adapters.

8 S6. Independent MV-Adapter audit

The Exact 76-object audit records 11 evaluated conditions/schedules and separates high-scale 1.50 and high-scale 1.00 families. The selected-pair CAI rule is `undefined_set_valued`. The LHL direct transfer has no clear practical gain over fixed-low or fixed mean; confidence intervals crossing zero are not interpreted as equivalence. Official pretraining UID disjointness is unknown. Absolute PSNR and other scores are not compared with the main MVPainter-style backbone. The full evidence is in `MV_ADAPTER_UNIFIED_RESULTS.csv` and `MV_ADAPTER_PAIRED_COMPARISONS.json`. The original `results/holdout_exact_76/PROTOCOL.json` is absent; `run_config.json` and `paired_bootstrap_exact.json` provide the available partial provenance.

9 S7. FAC controlled re-examination

FAC is a separate learned extension protocol, not a main contribution. The base model, VAE, and adapter are frozen; only lightweight controllers are trained on a disjoint training pool. Under the paired re-examination,

no trained configuration improves the training-free schedule. The paired foreground-PSNR deficits range from -1.10 to -2.94 dB, while the untrained exact-envelope control has $\Delta = -0.18$ dB with 95% CI $[-0.42, +0.05]$. The interval crossing zero is not an equivalence claim. The complete dose–response records are retained as a negative extension audit; the earlier positive FAC comparison used a different protocol and is not used in the revised claims.

10 S8. Bounded residual-budget controller pilot

We tested one residual-budget controller extension (TRB-TCAS) on the 24-object clean-v2 development probe before considering any holdout run. The run used 50 denoising steps, six unique target views, seed 42, shared per-object batches and initial latents, and fixed-low, C3/LHL, HLL and LLH controls. FG-LPIPS was the pre-specified primary metric. TRB obtained FG-LPIPS 0.1349, compared with 0.1404 for fixed-low and 0.1336 for C3; its PSNR was 22.700 dB, compared with 22.261 dB for fixed-low and 22.971 dB for C3. The calibration pass approximately doubled the mean controller time per object (12.342 s versus 6.075 s for C3). Since TRB did not improve the existing C3 schedule and added cost, the branch was stopped before a holdout run. These results are retained as a negative method-iteration record, not as a claim about the main TCAS method.

The machine-readable output is `TRB_DEV24_GATE_20260929.md` and the independent pilot `pilot_results.json`. The pilot does not justify relabelling LLH as a CAI-selected schedule: LLH is reported only as a fixed schedule control.

11 S9. Shared-input layer-wise ablation

The development ablation used 24 clean-v2 probe objects and seeds 42, 43, and 44. Each object was collated once per seed. All six methods reused the same target tensor, geometric input/features, and initial latent; the generation seed was reset before each method, including the stochastic condition-VAE encoding. The hash rows are in `coordination/layer_lhl_v1/shared_input_hashes.json`; raw rows, manifests, and SHA-256 sums are stored alongside it.

Table 7: Shared-input development ablation. Means are over 72 object-seed pairs. Lower is better for LPIPS.

Method	Full-PSNR	Full-SSIM	Full-LPIPS	FG-PSNR	FG-SSIM	FG-LPIPS	Edge-SSIM
Fixed-low	15.085	0.8235	0.2922	8.125	0.3454	0.2195	0.4896
C3/LHL	16.058	0.8299	0.2748	9.556	0.3929	0.2072	0.5063
Layer-fixed-low	16.861	0.8274	0.3501	10.753	0.4249	0.1964	0.5238
Layer-fixed-mean	16.892	0.8316	0.2796	11.179	0.4352	0.1889	0.5287
Layer-LHL	17.036	0.8315	0.3143	11.209	0.4352	0.1928	0.5239
Layer-LLH	18.165	0.8378	0.2562	12.364	0.4467	0.1785	0.5438

The paired layer-LHL minus fixed-low improvements are +1.951 dB Full-PSNR ([+1.624,+2.277]), +3.083 dB FG-PSNR ([+2.745,+3.387]), +0.0899 FG-SSIM ([+0.0802,+0.0992]), and −0.0267 FG-LPIPS ([−0.0295, −0.0238]). The layer-LHL minus layer-fixed-mean FG-LPIPS difference is +0.00388 ([+0.00273, +0.00518]), favoring layer-fixed-mean because lower is better. Layer-LLH minus layer-LHL is −0.01436 ([−0.01642, −0.01224]) on FG-LPIPS and is also better on the other six reported metrics. These intervals are paired percentile bootstrap intervals with 10,000 resamples and seed 20260929; they are not equivalence tests.

The shallow group uses the implementation cap of 0.8; the largest layer-LHL shallow value is 0.75, so no candidate value is clipped. This ablation supports layer-wise adapter scaling, but it does not establish a unique temporal optimum.

Runner sensitivity of the probe. The reference conditioning image undergoes a random stretch/compress (scale 0.5–1.5×) inside the dataset loader; targets, masks, depths, and view order are deterministic. Consequently all methods within one run are strictly paired, but absolute values are not comparable across separate runner invocations. A second, independent shared-input probe run of the same schedules (conducted for the complete factorial below) reproduces the LLH-over-LHL direction but changes the sign of the small LHL-vs-constant-low difference; we therefore claim only directions that hold in both runs and on the holdout.

12 S10. Complete binary temporal-pattern factorial (layer-wise)

The complete enumeration uses the same checkpoint, probe objects, seeds (42, 43, 44), unique6 views, 50 Euler steps, shared per-object inputs, and metric path as S9. The two levels are low = (deep 1.25, middle 1.25, shallow 0.50) and high = (deep 2.50, middle 2.50, shallow 0.75); each pattern assigns one level to each of the 17/16/17 early/middle/late step segments. The protocol manifest records that holdout selection is forbidden (development-only surface). Raw rows, manifests, and SHA-256 sums are in `coordination/layer_factorial_v1_20260930/`.

Table 8: Complete binary layer-wise factorial; means over 72 object–seed pairs. Lower is better for LPIPS.

Pattern	Full-PSNR	Full-SSIM	Full-LPIPS	FG-PSNR	FG-SSIM	FG-LPIPS	Edge-SSIM
LLL	20.801	0.8759	0.2200	16.020	0.6112	0.1124	0.5942
LLH	21.856	0.8832	0.1805	17.466	0.6219	0.1046	0.6131
LHL	20.561	0.8784	0.2008	15.661	0.6038	0.1193	0.5887
LHH	21.413	0.8837	0.1734	17.171	0.6194	0.1093	0.6113
HLL	18.433	0.8636	0.2246	14.082	0.5951	0.1295	0.5849
HLH	19.321	0.8721	0.1860	15.382	0.6132	0.1206	0.6089
HHL	18.155	0.8675	0.2092	13.747	0.5894	0.1355	0.5796
HHH	18.828	0.8731	0.1828	15.064	0.6114	0.1248	0.6073

Paired bootstrap comparisons on FG-LPIPS (first minus second; positive means the first is worse because lower is better): LHL−LLH +0.01472 ([+0.01168, +0.01864]), LHL−LHH +0.01008 ([+0.00799, +0.01226]), LHL−LLL +0.00693 ([+0.00419, +0.01076]), LHL−HLL −0.01012 ([−0.01286, −0.00736]), LHL−HLH −0.00123 ([−0.00401, +0.00168]), LHL−HHL −0.01612 ([−0.01940, −0.01283]), LHL−HHH −0.00547 ([−0.00847, −0.00221]). LLH is the strongest pattern on five of seven metrics and LHH on the remaining two (Full-SSIM, Full-LPIPS); no pattern dominates every metric.

13 S11. Strict-276 confirmation of pre-registered layer-wise candidates

The confirmation set re-runs layer-LLH, layer-fixed-mean, and a layer-LHL replica in one seeded runner (276 strict-holdout objects, unique6 views, 50 Euler steps, seed 42; per-object seeds fixed before dataset access, the initial latent, and the condition-VAE path). The protocol was frozen in `coordination/revision_next_20260930/EXPERIMENT_PRO` before any of these holdout numbers existed; candidates were selected only on probe FG-LPIPS. Raw rows, CSVs, manifests, and the analysis JSON are in `mvpainter-layer-confirmation-20260930/` (copies in `coordination/layer_confirmation_20260930/`).

Table 9: Strict-276 same-runner confirmation means.

Schedule	Full-PSNR	Full-SSIM	Full-LPIPS	FG-PSNR	FG-SSIM	FG-LPIPS	Edge-SSIM
Layer-fixed-mean	19.516	0.8503	0.2082	13.100	0.4550	0.1698	0.5363
Layer-LHL (replica)	19.635	0.8486	0.2278	12.974	0.4445	0.1719	0.5273
Layer-LLH	20.828	0.8514	0.1988	13.975	0.4620	0.1610	0.5476

Paired object-level 95% bootstrap intervals (10,000 resamples, seed 20260930; positive values favor the first schedule under its metric direction):

- **LLH − fixed-mean:** all seven metrics favor LLH (Full-PSNR +1.311 [+1.219, +1.400], 275/276; Full-SSIM +0.00114 [+0.00062, +0.00164]; Full-LPIPS −0.00936; FG-PSNR +0.875 [+0.769, +0.982], 240/276; FG-SSIM +0.00701; FG-LPIPS −0.00889 [−0.0097, −0.0081], 258/276; Edge-SSIM +0.01133).
- **LLH − layer-LHL replica:** all seven metrics favor LLH (Full-PSNR +1.193 [+1.123, +1.262], 276/276; Full-SSIM +0.00280; Full-LPIPS −0.02897; FG-PSNR +1.001 [+0.907, +1.097], 261/276; FG-SSIM +0.01744; FG-LPIPS −0.01097 [−0.0118, −0.0101], 264/276; Edge-SSIM +0.02031).

- **Fixed-mean − layer-LHL replica:** mixed. Fixed-mean is better on FG-LPIPS (+0.00208 [+0.0017, +0.0025], 202/276), FG-PSNR (+0.126), Full-SSIM (+0.00166), Full-LPIPS (−0.01961), FG-SSIM (+0.01043), and Edge-SSIM (+0.00897); the replica is better on Full-PSNR (+0.119 [+0.084, +0.153]).

Cross-runner replica check. The archived layer-LHL record used an unseeded Python RNG for the reference preprocessing draw and therefore has different draws from the replica. Their object-level Pearson correlation on FG-PSNR is 0.66 ($n = 276$); the replica's mean FG-PSNR is 12.97 versus the archived 14.78. Both records are retained; the archived record is a single unseeded run whose absolute values do not reproduce on restart, and all paired conclusions above come from the seeded same-runner set.

These results confirm the development-probe direction for the two promoted candidates on the holdout. They do not re-open schedule selection, do not compare absolute values across runners, and do not establish equivalence for any interval that crosses zero.

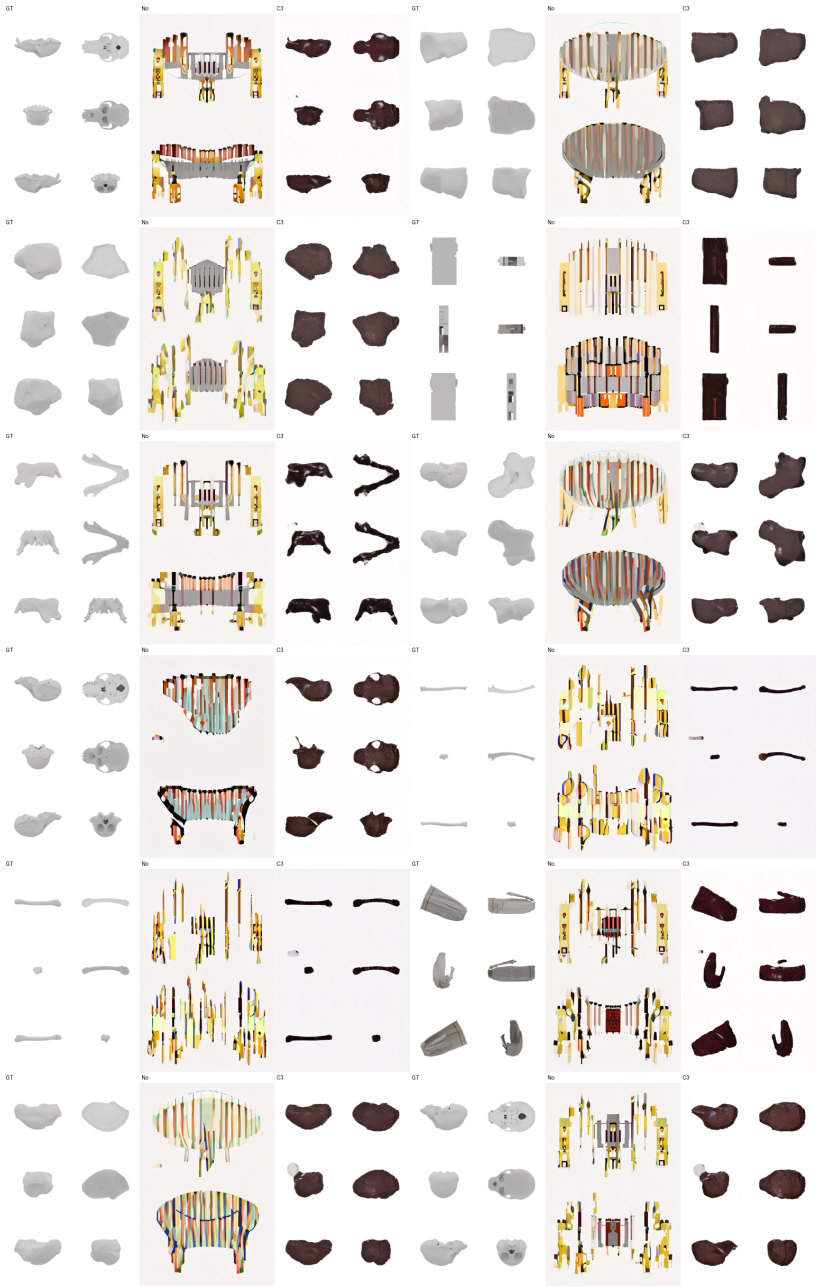


Figure 1: Complete 12-object comparison contact sheet. This is the full qualitative cohort, not a success-only selection; object-level baking metrics and coverage are reported separately in S4.